

Network Traffic Analysis Report

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1 Overview

This report presents an analysis of network traffic logs to identify patterns, anomalies, and potential security threats. The dataset contains over 2 million flows with attributes such as source/destination IPs, protocols, packet counts, and timestamps.

2 Part 1: Basic Network Statistics

2.1 Traffic Overview

- Total flows: 2,071,657
- Unique IPs: 34,801

2.2 Protocol Distribution

TCP dominates the traffic, followed by UDP and ICMP, indicating typical network behavior.

2.3 Top Communicating Entities

Top source and destination IPs show repeated communication patterns, likely representing servers or frequent connections.

2.4 Packet Analysis

- Average packet size: ~ 736 bytes
- High variance observed, indicating mixed traffic types

2.5 Traffic Patterns

- Two major traffic spikes detected using the 3σ rule
- Consistent IPs show stable communication patterns
- High-variance IPs indicate bursty or irregular activity

3 Part 2: Sublinear Traffic Estimation

3.1 Unique IP Estimation (HyperLogLog)

- Exact count: 34,801
- Estimated count: 35,724
- Error: 2.65%

The HyperLogLog algorithm achieves high accuracy using significantly reduced memory.

3.2 Heavy Hitters (Count-Min Sketch)

The most frequent destination IPs were accurately approximated with minimal error, demonstrating the effectiveness of Count-Min Sketch.

3.3 Membership Testing (Bloom Filter)

- False positives observed: 1%

Bloom filters provide efficient membership testing with acceptable probabilistic error.

4 Part 3: Anomaly Detection

4.1 Statistical Anomalies

Two anomalous time windows were detected using Z-score analysis, corresponding to significant traffic spikes.

4.2 Behavioral Analysis

- Certain IPs exhibit highly bursty behavior (e.g., 192.168.1.105)
- Sudden changes in activity suggest possible compromised hosts

4.3 Correlation Patterns

- Multiple sources targeting a single destination indicate possible DDoS behavior
- Single sources contacting many destinations suggest scanning activity
- Multi-protocol usage observed in short intervals

5 Part 4: Threat Analysis

5.1 Port Scanning

Several IPs contacted a large number of destination ports (e.g., >1000), strongly indicating port scanning activity.

5.2 DDoS Behavior

A destination IP receiving traffic from hundreds of distinct sources in short intervals suggests a distributed denial-of-service (DDoS) pattern.

5.3 IP Hopping

Some sources communicate with a large number of destinations and ports, indicating evasive or automated behavior.

5.4 Payload Analysis

Due to sparse payload data, anomalies were detected using payload size. Extremely large payloads may indicate data exfiltration.

5.5 Command-and-Control (C2)

Repeated communication between specific source-destination pairs suggests potential command-and-control behavior.

5.6 Risk Classification

IPs were classified into risk categories based on multiple indicators:

- High Risk: Multiple attack patterns detected
- Medium Risk: Moderate anomalies observed
- Low Risk: Normal behavior

6 Summary

- Traffic is largely stable with occasional spikes
- Sublinear algorithms provide accurate estimates with low memory
- Multiple anomalies align with potential attack patterns
- Evidence of port scanning, DDoS, and coordinated behavior observed

7 Conclusion

The analysis demonstrates the effectiveness of combining statistical methods, sublinear algorithms, and behavioral analysis to detect network anomalies and security threats. The identified patterns indicate the presence of potentially malicious activity, emphasizing the importance of continuous monitoring.